

MARIA PAZ MALUENDA WALTON

Software Engineer

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SUMMARY

Software engineer with an M.S. in Computer Science and Engineering and experience across full-stack applications, computer vision, robotics research, and augmented reality. Builds practical, human-centered systems with Python, JavaScript/TypeScript, FastAPI, Django, React, PostgreSQL, YOLO, OpenCV, ONNX Runtime, Unity, and ROS.

SKILLS

Languages: Python, C++, JavaScript, TypeScript, SQL, HTML, CSS

Frameworks & Tools: React, FastAPI, Django, PostgreSQL, Tailwind CSS, Git, Docker, REST APIs

AI / Robotics / XR: YOLO, OpenCV, ONNX Runtime, Hugging Face, CVAT, Unity, ROS, Human-Robot Interaction, AR/VR

Systems: Algorithms, Data Structures, Operating Systems, Linux, Digital Forensics, SQLite, ADB

PROJECTS

GameMatch | Computer Vision, AR & Full-Stack Platform | 2025-2026

- Designed and built an end-to-end board game recognition platform spanning AR, computer vision, and web workflows.
- Developed a FastAPI backend supporting webcam and image-based recognition workflows.
- Trained a custom YOLO object detection model on a self-curated CVAT-annotated dataset.
- Evaluated OCR, image embeddings, and object detection pipelines to identify the most reliable recognition approach.
- Integrated ONNX Runtime inference and BoardGameGeek metadata to recognize games and surface recommendations.

Human-Robot Interaction Trust Repair Research | University of Nevada, Reno | 2025-2026

- Contributed to research on trust repair in child-robot collaboration using a NAO humanoid robot.
- Supported experimental design, participant studies, survey development, and research methodology.
- Evaluated apology and restitution strategies following robot failures.

GeekGuide | React, Django, Python, PostgreSQL, Tailwind CSS

- Built a full-stack board game discovery application using React, Django, Python, PostgreSQL, and Tailwind CSS.
- Developed search and filtering functionality with a responsive user interface for discovering board game information.

Aero Hand | Python, ROS

- Configured and debugged a tendon-driven robotic hand performing American Sign Language gestures using Python and ROS.
- Investigated robotic motion, calibration, and software integration challenges across hardware and software components.

EXPERIENCE

Graduate Teaching Assistant | University of Nevada, Reno | 2024-2026

CS135 - Computer Science I | CS302 - Data Structures

- Supported instruction in introductory programming and data structures.
- Mentored students in debugging, algorithms, object-oriented programming, and software engineering fundamentals.

Architectural Intern / Junior Designer | Skidmore, Owings & Merrill (SOM), San Francisco, CA

Sep 2018-Mar 2019 | Feb 2021-May 2022

- Developed computational design tools that streamlined architectural exploration and iterative workflows.
- Collaborated with multidisciplinary engineering and design teams on large-scale projects.
- Applied automation and software development to improve design efficiency and decision-making.

EDUCATION

University of Nevada, Reno | M.S. Computer Science and Engineering | May 2026

General Assembly | Software Engineering Immersive | 2023

University of Chile | B.S. Architecture | 2020

Relevant Coursework: Algorithms, Operating Systems, Robotics, Human-Robot Interaction, Augmented & Virtual Reality, Full-Stack Web Development, Digital Forensics